

Brandon Dominique

Boston, MA

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PhD candidate at Northeastern University researching interpretability and uncertainty quantification for trustworthy machine learning, applied to healthcare. Developing post-hoc methods for model reliability, using calibration, conformal prediction, and clustering to discover groups where model performance differs. Prior research at IBM on model transparency and red teaming LLMs for safer deployment. Focused on building interpretability tools that give users and domain experts enough insight to trust model predictions in high-stakes settings.

SELECTED EXPERIENCE

Northeastern University, Graduate Research Assistant

Sep 2020-Current

Advisor: Jennifer Dy

- Developed SCC, a method that discovers semantically coherent groups where vision language models are miscalibrated, reducing expected calibration error by 79% (Med-Reasoner Workshop, NeurIPS 2026)
- Developed an interpretable benchmark of deep learning models for skin cancer detection, revealing intersectional group-level performance disparities (MELBA, 2025)
- Developing group-conditional conformal prediction algorithms that produce tighter prediction sets with guaranteed coverage across discovered groups
- Collaborating with clinicians at Memorial Sloan Kettering Cancer Center to develop reliable deep learning methods for dermatology, using uncertainty-based clustering and multimodal LLMs

IBM, Research Intern

May 2021-Aug 2023 (Three Summers)

- Worked within IBM's Responsible AI team, collaborating with researchers focused on transparency, fairness, and trustworthy AI for enterprise deployment
- Built a web-based interface giving each stakeholder role tailored access to model documentation, including training methodology, hardware performance, and architecture, across multiple analog AI models (Best Student Paper Award, IEEE SSE 2023)
- Designed and evaluated prompt templates to help non-expert users audit LLMs for biased outputs; a 29-participant study showed the templates improved users' auditing performance (HEAL Workshop, CHI 2024)
- Conducted semi-structured interviews with participants across multiple stakeholder roles to identify transparency needs for hardware-aware AI models

EDUCATION

Northeastern University, Boston, MA

Ongoing

Ph.D. Computer Engineering

Advisor: Jennifer Dy

Graduate Student Ambassador, Computer Engineering Department and GEM Fellowship

Northeastern University, Boston, MA

May 2023

M.S. Computer Engineering

Graduate Student Ambassador, Computer Engineering Department and GEM Fellowship

New Jersey Institute of Technology, Newark, NJ

May 2020

B.S. Computer Engineering, *Magna Cum Laude*

SKILLS

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- **Research:** Interpretable ML, Post-Hoc Explainability, Uncertainty Quantification
 - **Languages/Frameworks:** Python, Tensorflow, PyTorch, pandas, NumPy, scikit-learn, SLURM
 - **Reviewer:** ICML, AAAI
 - **Other:** Cross-Domain Collaboration, Stakeholder Communication, Mentorship

PAPERS

Dominique B., Lam P., Torop M., Kurtansky N., Weber J., Kose K., Rotemberg V., Dy J. (2026). Discovering Bias by Learning Semantically Coherent Calibration Groups. *Med-Reasoner Workshop, NeurIPS 2026. Extended version under review.*

Dominique B., Lam P., Dy J. (2026). Uncertainty-Aware Group-Conditional Conformal Prediction. *Under review.*

Dominique B., Lam P., Kurtansky N., Weber J., Kose K., Rotemberg V., Dy J. (2025). On the Role of Calibration in Benchmarking Algorithmic Fairness for Skin Cancer Detection. *In the Journal of Machine Learning for Biomedical Imaging (MELBA).*

Dominique B., Piorkowski D., Nagireddy M., Baldini I. (2024). Prompt Templates: A Methodology for Improving Manual Red Teaming Performance. *In 1st HEAL Workshop at CHI Conference on Human Factors in Computing Systems.*

Dominique B., Piorkowski D., El-Maghraoui K., Herger M. (2023). FactSheets for Hardware-Aware AI Models: A Case Study of Analog In Memory Computing AI Models. *In IEEE International Conference on Software Services Engineering.* (Best Student Paper Award)

FELLOWSHIPS, AWARDS, AND TALKS

ICML 2026 Gold Reviewer	April 2026
Invited Talk , "Post-Hoc Methods for Detecting and Correcting Subgroup Bias in Medical AI Systems", University College London, Alan Turing Institute	April 2026
GEM Fellowship	August 2021
LSAMP STARS Fellowship	August 2020